# Band-1 Hadamard Order 4372 — Miyamoto (Paley-I seed H_1092); Gap 2052 Still OPEN

H_1092/C_1094/H_4372 PASS; 4372 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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**Document class.** Scientific computational / construction certificate (Band-1).
**Program.** Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
**Parents.** `2026-09-15_Band1_Hadamard_4196_Miyamoto`; `2026-09-15_Band1_Hadamard_4132_Miyamoto` (direct Paley-I seed pattern).
**Lab.** `/workspace/experiments/band1/hadamard_4372_miyamoto/`.
**Cutoff.** 2026-09-15T15:26:09Z (~11:26AM ET).

**Honesty stamp (non-negotiable).**
This memo claims **finite constructions of orders 1092 and 4372 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I seed $H_{1092}$ ($q = 1091$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4372 is a **Miyamoto-only bonus (NOT in SPK)**: Paley-I **FAIL** ($q = 4371 = 3 \cdot 31 \cdot 47$ not prime); Paley-II **FAIL** ($q = 2185 = 5 \cdot 19 \cdot 23$ not prime); Kronecker **FAIL** ($4372 = 4 \cdot 1093$, 1093 not Hadamard-admissible). Paley-II path for seed $H_{1092}$ **SKIP** ($q = 545 = 5 \cdot 109$ not prime). `clay_closed = false`.

## Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4196 on box. This increment ships **order 4372** ($4372 = 4 \cdot 1093$, 1093 prime $\equiv 1 \pmod 4$) via Miyamoto with a direct **Paley-I seed** $H_{1092}$ ($q = 1091$). Symmetric Paley conference $C_{1094}$ over prime 1093; $H_{4372} = \text{Miyamoto}(H_{1092}, C_{1094})$. Unit tests **37/37 PASS**. SPK next gap remains **2052** (OPEN). Order 4372 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

**Keywords.** Hadamard matrix; order 4372; Miyamoto; Paley type I; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

## 1. Statement lock

**Definition.** A Hadamard matrix of order $n$ is an $n \times n$ matrix $H$ with entries in $\pm 1$ satisfying $HH^\top = nI$.

**Finite lemmas (this increment).** 1. $H_{1092}$ constructed and verified (Paley type I, $q = 1091$). 2. $C_{1094}$ constructed and verified (Paley conference, $q = 1093$): $CC^\top = 1093I$, zero diagonal. 3. $H_{4372}$ constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. SPK: $4372 \notin$ SPK (Miyamoto-only). Paley-I/II/Kronecker paths for 4372 **rejected**. Next constructive gap remains **2052** (OPEN).

## 2. Arithmetic and eligibility

| Quantity | Value |
|---|---|
| Target order | $4372 = 4 \cdot 1093$ |
| $q = 1093$ | prime, $1093 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible |
| Needed $H_{q-1}$ | $H_{1092}$ |
| Needed conference | order $1094 = q + 1$ |
| $H_{1092}$ path (used) | Paley type I, $q = 1091$ (prime $\equiv 3 \pmod 4$); $q + 1 = 1092$ |
| $H_{1092}$ Paley-II | **SKIP**: $2(q + 1) = 1092 \Rightarrow q = 545 = 5 \cdot 109$ not prime |
| Conference path | Symmetric Paley over prime 1093 |
| Paley-I for 4372 | **REJECTED**: $q' = 4371 = 3 \cdot 31 \cdot 47$ not prime |
| Paley-II for 4372 | **REJECTED**: $q' = 2185 = 5 \cdot 19 \cdot 23$ not prime |
| Kronecker for 4372 | **REJECTED**: only $4 \times 1093$; $1093 \not\equiv 0 \pmod 4$ (not admissible) |

## 3. Construction results

| Object | Method | $\max \lvert HH^\top - nI \rvert$ (or conf) | Wall (s) | Status |
|---|---|---|---|---|
| $H_{1092}$ | Paley type I, $q = 1091$ | 0 | 1.1476 | **PASS** |
| $C_{1094}$ | Paley conference, $q = 1093$ | 0 | 1.105 | **PASS** |
| $H_{4372}$ | Miyamoto($H_{1092}, C_{1094}$) | 0 | 37.1819 | **PASS** |

Primary pipeline wall (incl. unit tests + SPK refresh) $\approx 40.8278$ s. Compressed $\pm 1$ matrices saved as `H1092_pm1.npz`, `C1094_pm1.npz`, `H4372_pm1.npz` (int8).

## 4. SPK closure note

Seeds: all admissible $\leq 2000$ $\cup$ on-box prior classical/SPK ships (through 4036/3908/3748/… + **1020** Paley-II + **1032** Paley-I + **524/1048** Double Paley-I; **exclude** 4196/4132/4084/3716/3524/3284/3076/3044/ Miyamoto-only) $\cup$ verified $H_{1092}$ (Paley-I) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4500). **Miyamoto is not folded into SPK**; order 4372 remains a Miyamoto-only bonus.

| Quantity | Value |
|---|---|
| Known count | 964 |
| 4372 in SPK | **False** (Miyamoto-only) |
| 1092 in SPK | **True** (Paley-I) |
| Next constructive gap | **2052** (OPEN) |
| Miyamoto-only bonus? | **Yes** (like 4196/4132/4084/3716/3524/…) |

## 5. Gap 2052 (OPEN; method triage / non-finding ONLY)

| Path | Result |
|---|---|
| Paley-I ($q = 2051$) | FAIL |
| Paley-II ($q = 1025$) | FAIL |
| Kronecker | FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$) |
| Miyamoto + Paley conference ($q = 513$) | FAIL (513 not prime) |
| GS / SDS / Williamson / Yamada | lit-only; not constructed this run |

**Status: OPEN.** Non-finding $\neq$ counterexample. Conjecture remains OPEN.

## 6. Unit tests

**37/37 PASS** (Sylvester/Paley/Miyamoto controls; $q = 1091/1093$ arithmetic; Paley-I seed for $H_{1092}$; Paley-II skip; SPK path rejection for 4372; Miyamoto-only arith guard; onbox seeds include 504/524/1020/1032/1048 / exclude 4084/4132/4196/4372; reject-all-ones; conference control; gap-2052-not-filled guard).

## 7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented $J$.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 4372 is a **Miyamoto-only bonus (NOT in SPK)**.

## 8. Next optional

method-attack SPK gap **2052** OR next Miyamoto ($q = 1097 \to n = 4388$ Double(PaleyI half=548); or $q = 1117 \to n = 4468$ PaleyII($q = 557$); or $q = 1153 \to n = 4612$ PaleyI($q = 1151$)) OR Track B privacy / Object 1 meter await.

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*Lab:* `/workspace/experiments/band1/hadamard_4372_miyamoto/` · *Script:* `verify_hadamard_4372.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.